MIT 524 · Cloud and Emerging Computing Paradigm

UPTIME QUEST

MANUAL

Uptime Quest Manual

A beginner's guide to our simulation

 HA ZONE

 FT ZONE

 DR ZONE

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QUICK START (60 SECONDS)

1. Open Uptime Quest and press ENTER on the title screen (or tap PRESS ENTER).
2. On the zone screen, pick HA, FT or DR with the arrow keys and press ENTER, or just click a zone.
3. Sound is off at first. Click BGM for music and SFX for sound effects (top-right corner) if you want them.
4. Each zone has a short instruction box at the top. Follow it, break things, and watch what happens.
5. Press ESC (or the back arrow) to return to the zone screen at any time.

HOW TO READ THIS MANUAL

Each zone chapter has the same parts: **the goal**, **a screen tour** with numbered labels, **how it works**, and a **step-by-step walkthrough**. If you only have a few minutes, read the Quick Start above and the walkthrough of the zone you want to play.

Numbered yellow badges on the screenshots match the numbered lists right under them.

1 WELCOME TO UPTIME QUEST

Uptime Quest is an 8-bit simulation that goes with the report *Cloud Reliability, Fault Tolerance and Disaster Recovery*. Instead of only reading about how cloud systems stay online, you get to break them on purpose and watch them recover.

The three zones

The report explains cloud reliability as three layers of defense. Each layer is one zone in the game:



HA ZONE

PREVENT

High Availability keeps everyday failures from causing downtime. You crash servers and watch a load balancer route around them.



FT ZONE

SURVIVE

Fault Tolerance lets a system survive failures that do happen. You compare acting before a failure (proactive) with acting after it (reactive).



DR ZONE

RECOVER

Disaster Recovery brings a business back after a large-scale disaster. You pick a recovery strategy, trigger a disaster and race the clock.

Meet Andy Lim



Andy Lim owns a mid-sized business that sells light bulbs and electrical fixtures, both retail and wholesale. He moved his store to the cloud, he has a small IT team and a limited budget, and his holiday sale is coming up. Every zone is about keeping **Andy's store** online. Andy is a fictional case study made for this report.

⚠ WHAT'S REAL AND WHAT'S ILLUSTRATIVE

The **concepts** are real and come from the report's sources: redundancy models, health checks and failover, proactive and reactive techniques, the fault → error → failure chain, RTO and RPO, and the four AWS recovery strategies.

The **numbers inside the games** (seconds, request counts, order counts, costs) are made up for the demo and sped up so a whole outage fits in a few seconds. They follow the right order (for example, Backup & Restore is always slower and cheaper than Active/Active), but they are not measurements.

2 GETTING STARTED

Opening the app

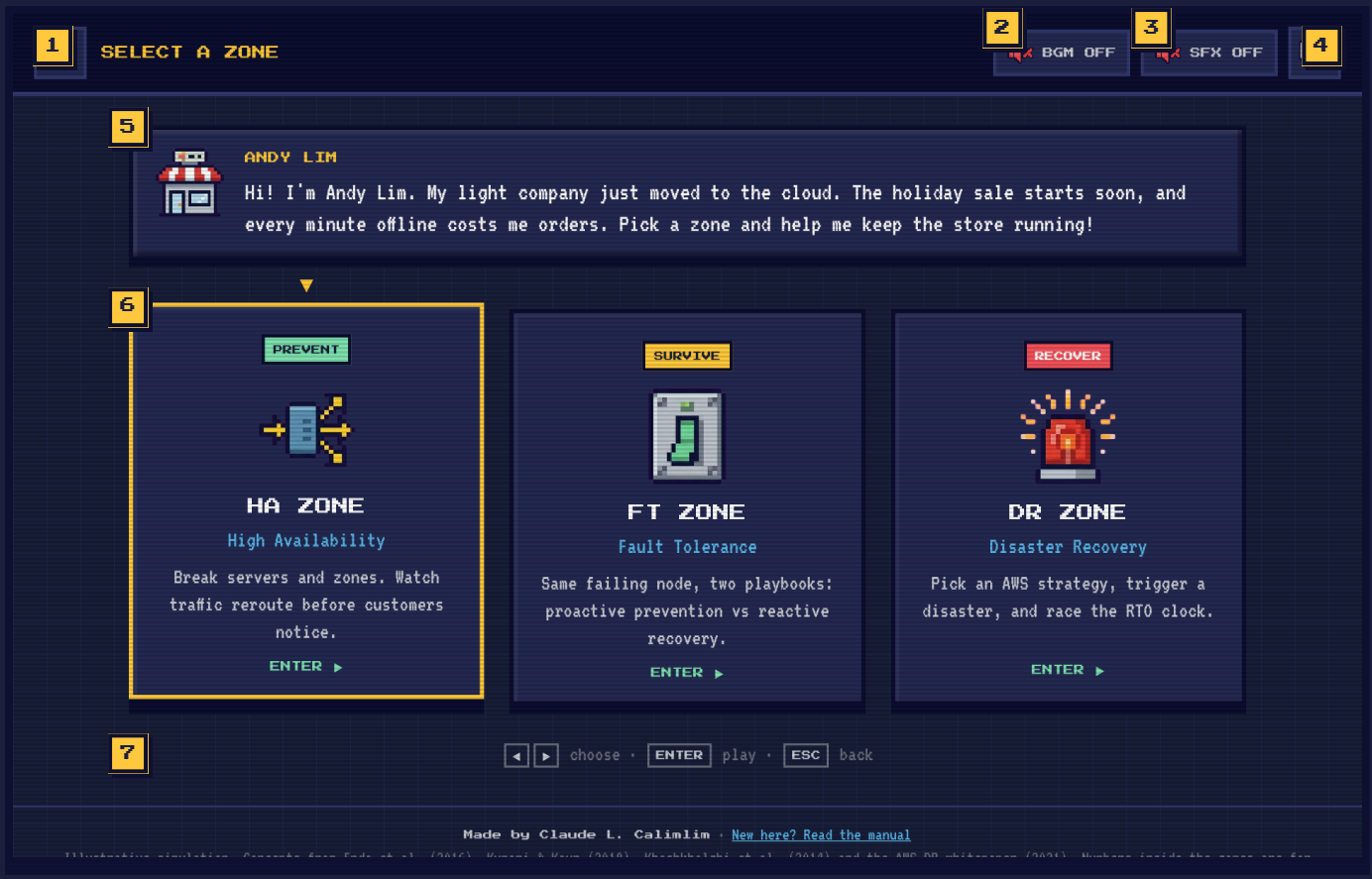
- **Online:** open the Uptime Quest link shared by the presenter in any modern browser (Chrome, Edge, Firefox or Safari).
- **Offline:** if you have the project folder, double-click `index.html`. It runs without internet.
- **Phones and tablets** work too. Tap instead of using the keyboard.

The title screen



1. The game title.
2. **PRESS ENTER:** press **ENTER** or **SPACE**, or click / tap this button, to continue.
3. **HOW TO PLAY:** opens this manual. It's also linked from the About box and the zone screen's footer. Inside the manual, the back arrow (or **ESC**, or your browser's Back button) returns you to the game.
4. The app's creator.
5. **BGM** and **SFX:** turn music and sound effects on or off before you start.

The zone screen (hub)



1. **Home**: back to the title screen (or press ESC).
2. **BGM**: background music on/off. It is **off** when you first open the app.
3. **SFX**: sound effects (beeps, alarms, clicks) on/off. Also **off** at first.
4. **About**: what the app is, its sources, credits, and a link to this manual.
5. **Andy Lim's message**: he types out a short intro when you arrive.
6. **Zone cards**: the yellow outline and arrow show which zone is selected. Move with ◂ ▸ (or ▴ ▾), hover with the mouse, then press ENTER or click to play.
7. A reminder of the keyboard controls (hidden on touch screens).

Sound

Uptime Quest starts silent: both **BGM** and **SFX** are **OFF** every time you open the app. Click **BGM** for the looping 8-bit music and **SFX** for game sounds.

- The **BGM** and **SFX** buttons are on the title screen, the zone screen, every zone, and the top bar of the online manual.
- Once you turn the music on, it keeps playing everywhere in the app, including while you read the online manual.
- Reloading the page turns both back off.

Jumping straight to a screen

Add one of these to the end of the app's address to skip the menus: `#hub` (zone screen), `#ha`, `#ft`, `#dr` or `#manual` (this manual). This works on a fresh load and on a tab that's already open.



On a phone

The layout stacks into one column. Everything is tap-to-use: tap a zone card to open it, tap the back arrow to leave. Keyboard hints are hidden on touch screens.

In the FT zone the two panels sit on top of each other instead of side by side, so scroll down to see the reactive panel.

3 CONTROLS CHEAT SHEET

Everything can be done with a mouse or by tapping. These keyboard shortcuts are optional.

WHERE	KEY	WHAT IT DOES
Title screen	ENTER / SPACE	Start (go to the zone screen)
Zone screen	◂ ▸ ▴ ▾	Move between zones
	ENTER	Open the selected zone
	ESC	Back to the title screen
Any zone	ESC	Back to the zone screen
HA zone	1 - 4	Pick a redundancy model (None, 2N, N+M, N-way active)
	C	Chaos mode on/off
	R	Repair all servers
FT zone	ENTER / SPACE	Start the scenario
DR zone	1 - 4	Pick a strategy (Backup & Restore, Pilot Light, Warm Standby, Active/Active)
	ENTER / SPACE	Trigger the disaster
About box	ESC	Close it (or click outside it)
Manual	ESC	Back to the game (arrow keys and space scroll the page as usual)

Mouse and touch

ACTION	RESULT
Click / tap a server (HA)	Crashes it. Click a crashed server again to repair it.
Click OUTAGE on a zone column (HA)	Crashes every server in that availability zone at once.
Click a technique button (FT)	Uses that technique. Buttons only work when they're lit up.
Click a strategy card (DR)	Picks it and shows what runs in the recovery region.
Click BGM / SFX	Turns music / sound effects on or off.

4 HA ZONE: HIGH AVAILABILITY

GOAL Keep Andy's store answering customers while servers fail, and see how each redundancy model handles it.

Screen tour



1	**Redundancy models**: four buttons (keys 1-4). Each has a real-world example under its name.
2	**Model description**: one line explaining the selected model.
3	**Customers**: they send about 6 requests every second (the small yellow squares).
4	**Load balancer**: sends each request to a healthy server, taking turns. The small monitor above it flashes on every **health check** (once per second).
5	**Availability zones 1-3**: separate data centers. Columns marked "unused" have no server in the current model.
6	**OUTAGE**: knocks out every server in that zone at once, like a power failure in one data center.
7	**Server tile**: click it to crash it, click again to repair it. It shows its name, its status (see table below) and a **load bar**.
8	**REPAIR ALL** (R): fixes every crashed server.
9	**CHAOS MODE** (C): crashes a random server every 3-5.5 seconds and repairs it 6 seconds later.
10	**RESET**: restarts the current model with fresh counters.

How it works

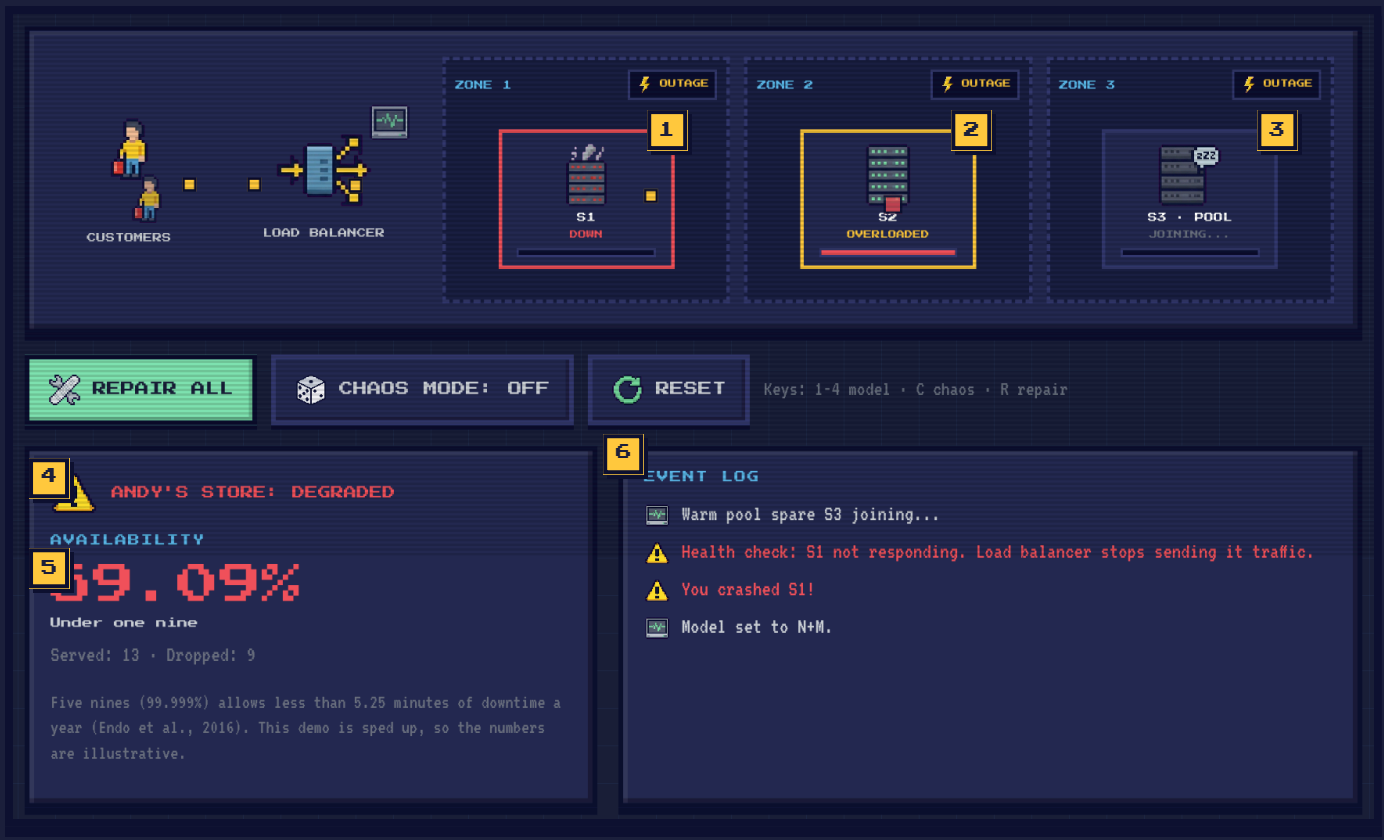
When a server crashes, the load balancer **doesn't know right away**. It keeps sending requests to the dead server until the next health check, up to 1 second later, and those requests are dropped (they flash red). After the health check, the load balancer stops using that server. What happens next depends on the model:

MODEL	SERVERS	WHEN ONE FAILS	REAL-WORLD EXAMPLE
No redundancy	1 server	Nothing takes over. The store is down until you repair it. This is a single point of failure.	-
2N	1 active + 1 full standby in another zone	After the health check, the standby is promoted to primary. That takes 1.5 more seconds, and requests are dropped in the meantime.	Amazon RDS Multi-AZ
N+M	N = 2 active + M = 1 spare in a warm pool	The surviving server is overloaded (it has to do the work of two) and about half the requests fail until the spare joins 1.5 seconds later.	EC2 Auto Scaling warm pools
N-way active	3 active, all serving	The other two absorb the work right away. They were only running at about 67% load, so nothing gets dropped after the health check. Lose two at once and the last one overloads.	Netflix multi-region active-active

Server status labels

LABEL	MEANING
SERVING	Healthy and receiving traffic.
CRASHED · LB UNAWARE	Dead, but the load balancer hasn't noticed yet. Requests sent here are lost.
DOWN	Dead, and the load balancer has stopped sending it traffic.
STANDBY / WARM POOL	A healthy backup that isn't serving yet.
JOINING...	A standby or spare taking over (1.5 seconds).
WAITING FOR CHECK	Repaired, waiting for the next health check before it gets traffic again.
OVERLOADED	Getting more work than it can handle, so some requests fail. Its load bar turns red.

Reading the results



1	A crashed server (DOWN).
2	The surviving server, OVERLOADED and dropping some requests.
3	The warm-pool spare, JOINING to restore capacity.
4	**Store status**, based on the last 3 seconds: **ONLINE** if more than 90% of requests succeeded, **DEGRADED** if 25–90%, **DOWN** if under 25%.
5	**Availability**: requests served ÷ all requests since the model was selected. Below it, the number of "nines" (99% is two nines, 99.9% is three).
6	**Event log**: the newest event is on top. It explains every crash, health check and failover.

When you switch models or leave the zone, your run is saved to the **YOUR RUNS PER MODEL** table at the bottom, as long as it had at least 20 requests. The table keeps your longest run for each model. Use it to compare models. It clears when the page is reloaded.

Walkthrough

MISSION 1 — A single point of failure

1. Press **1** (or click **NO REDUNDANCY**).
2. Click server **S1** to crash it.
3. Watch every request turn red and the store switch to **DOWN**. Nothing can take over.
4. Click S1 again to repair it. It waits for the next health check, then serving resumes.

MISSION 2 2N failover

1. Press **2**. S1 is active in Zone 1 and S2 is the standby in Zone 2.
2. Crash S1. Its label reads **CRASHED · LB UNAWARE** until the health check catches it.
3. Read the event log: "Promoting standby S2 to primary...", then "S2 is the new primary".
4. Repair S1. It comes back as the new **standby**, not the primary.

MISSION 3 N+M overload

1. Press **3**. S1 and S2 are active and S3 waits in the warm pool.
2. Crash S1. S2's load bar turns red (**OVERLOADED**) because it's doing the work of two.
3. After 1.5 seconds S3 joins and both bars go back to normal.

MISSION 4 N-way active under chaos

1. Press **4**, then press **C** to turn on chaos mode.
2. Let it run for 20-30 seconds. Servers crash and recover on their own.
3. Switch back to **2** with chaos still on and compare the availability numbers in the runs table.



Try this: in N-way active, use **OUTAGE** on Zone 2. One zone going down is survivable when traffic is spread across zones. That's why cloud providers recommend deploying to several availability zones.

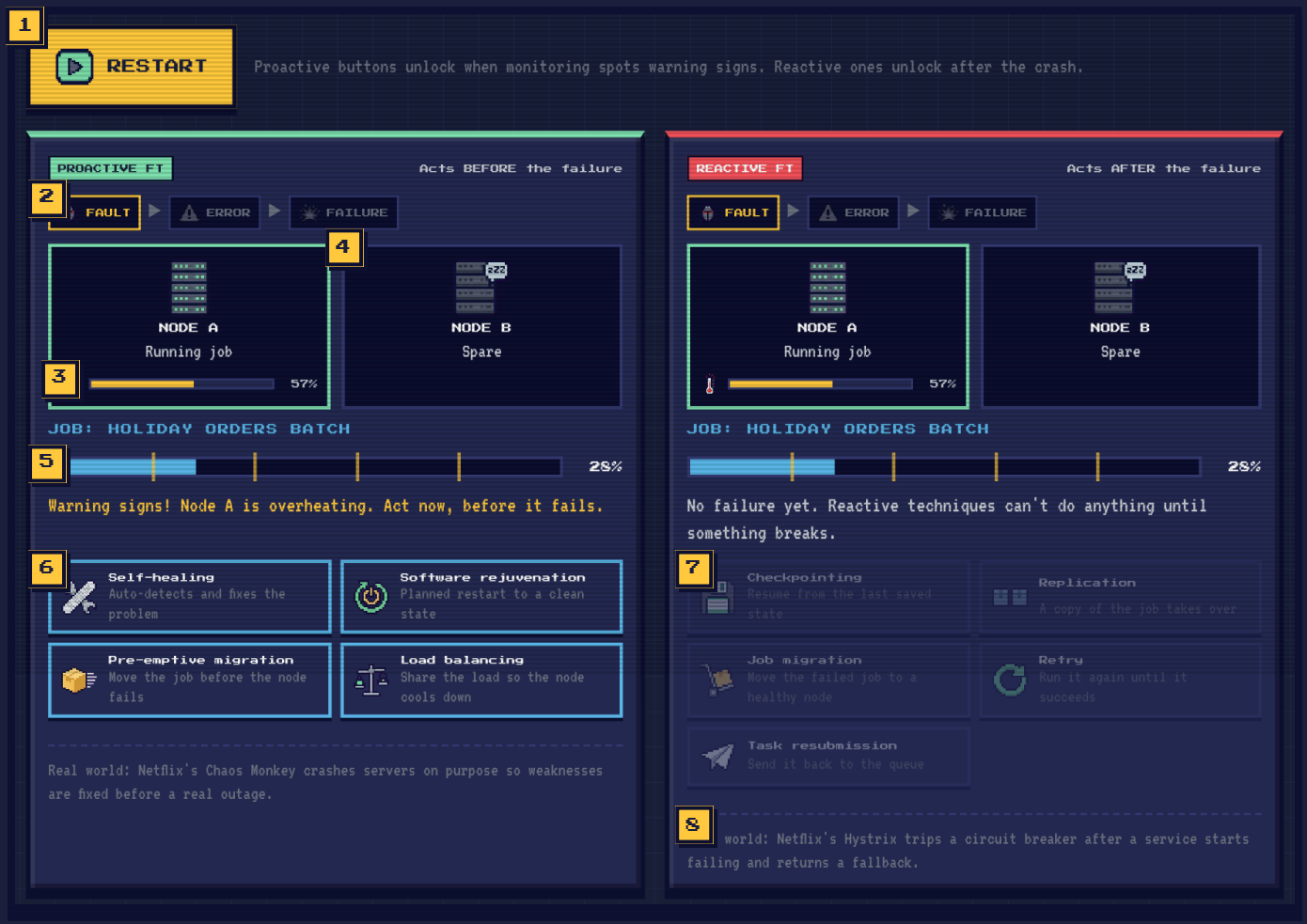
5 FT ZONE: FAULT TOLERANCE

GOAL Finish Andy's holiday orders job even though Node A is about to fail, once by preventing the failure and once by recovering from it.

The idea

Both panels show the **same situation at the same time**: Node A is running a job and slowly overheating. The **left panel (Proactive FT)** only lets you act **before** Node A fails. The **right panel (Reactive FT)** only lets you act **after** it fails. Comparing the two is the point of this zone.

Screen tour



1. START SCENARIO / RESTART (ENTER): starts both panels together.
2. **Fault → error → failure chain**: each step lights up as the problem gets worse. **Fault** = the underlying problem, **error** = the system's internal state goes wrong, **failure** = the user sees it.
3. **Node A temperature**: rises from 0% to 100% in about 8 seconds. At 30% the fault shows (warning signs), at 65% the error shows, at 100% Node A crashes.
4. **Node B**: a healthy spare that some techniques use.
5. **Job bar**: the holiday orders job takes about 16 seconds. The yellow lines are **checkpoints**: the job saves its progress at 20%, 40%, 60% and 80%.

6 **Proactive techniques:** unlock (they flash) once the temperature reaches 30% and stay available until Node A crashes. That gives you about 5–6 seconds.

7 **Reactive techniques:** locked until Node A crashes. Nothing has broken yet, so there's nothing to recover from.

8 A real-world example for each side: Netflix's Chaos Monkey (proactive) and Hystrix (reactive).

Proactive techniques (left panel)

TECHNIQUE	WHAT HAPPENS IN THE GAME	WORK LOST	EXTRA TIME
Self-healing	The system spots the problem and fixes Node A by itself. The temperature drops and the job continues.	0%	0.8 s
Software rejuvenation	A planned restart puts Node A back in a clean state, then the job picks up where it was.	0%	1.5 s
Pre-emptive migration	The job moves to Node B before Node A fails. Node A still crashes, but nothing was running on it, so there's no failure for the user.	0%	0.5 s
Load balancing	Node B takes half the work, Node A cools to about 45% and never crashes.	0%	0 s

If you don't pick a proactive technique in time, Node A crashes and the left panel shows **"Too late!"**. Proactive techniques can't help once the failure has happened. Press **RESTART** to try again.

Reactive techniques (right panel)

Node A crashes when the job is at about 49%. Then the reactive buttons unlock:

TECHNIQUE	WHAT HAPPENS IN THE GAME	WORK LOST	EXTRA TIME
Checkpointing	Node A restarts and the job resumes from the last checkpoint (40%), not from zero.	about 10%	1.4 s
Replication	A copy of the job was already running on Node B, so it takes over at once. It costs double the resources.	0%	0.3 s
Job migration	The failed job moves to healthy Node B and starts over there.	all (about 49%)	0.6 s
Retry	Node A restarts and runs the job again. The first attempt fails because the node is still unstable, and the second succeeds. The job starts over.	all (about 49%)	3.2 s
Task resubmission	The job goes back to the queue, waits for a free slot, then starts over on Node B.	all (about 49%)	1.6 s

"Extra time" counts the recovery pause only, not the time you take to choose.



1. The proactive panel when nothing was picked in time: **FAILURE NOT PREVENTED**.
2. The chain reaches **FAILURE** (red) once the user is affected.
3. The reactive panel tells you where the job stopped.
4. Reactive techniques are now unlocked and flashing. Pick one.

After a run



1. **Same failure, two playbooks**: the work lost on each side, side by side.
2. **The takeaway**: proactive tries to stop the failure, reactive limits the damage, and real systems use both.
3. **Techniques tried**: lights up each of the 9 techniques you've used. Try all 9 for a trophy. It resets when the page is reloaded.

Walkthrough

RUN 1 Prevent and recover

1. Click **START SCENARIO**.
2. Watch the temperature bar. At 30%, **FAULT** lights up and the proactive buttons start flashing.
3. On the left, click **Pre-emptive migration**. The job jumps to Node B.
4. On the right, wait for Node A to crash (about 8 seconds in), then click **Checkpointing**.
5. Wait for both jobs to finish and read the summary: 0% lost on the left, about 10% lost on the right.

RUN 2 Miss the window

1. Click **TRY OTHER TECHNIQUES** (or RESTART) and don't touch the left panel.
2. When Node A crashes, the left panel shows **Too late!** This is why proactive techniques need monitoring and early warning.
3. On the right, try **Job migration** and notice the job restarts from 0%.



Try this: compare **Checkpointing** with **Retry**. Both use Node A again, but only checkpointing keeps your progress.

6 DR ZONE: DISASTER RECOVERY

GOAL Pick a recovery strategy for Andy's store, survive a disaster in Region A, and compare downtime, data loss and cost.

Two numbers to watch

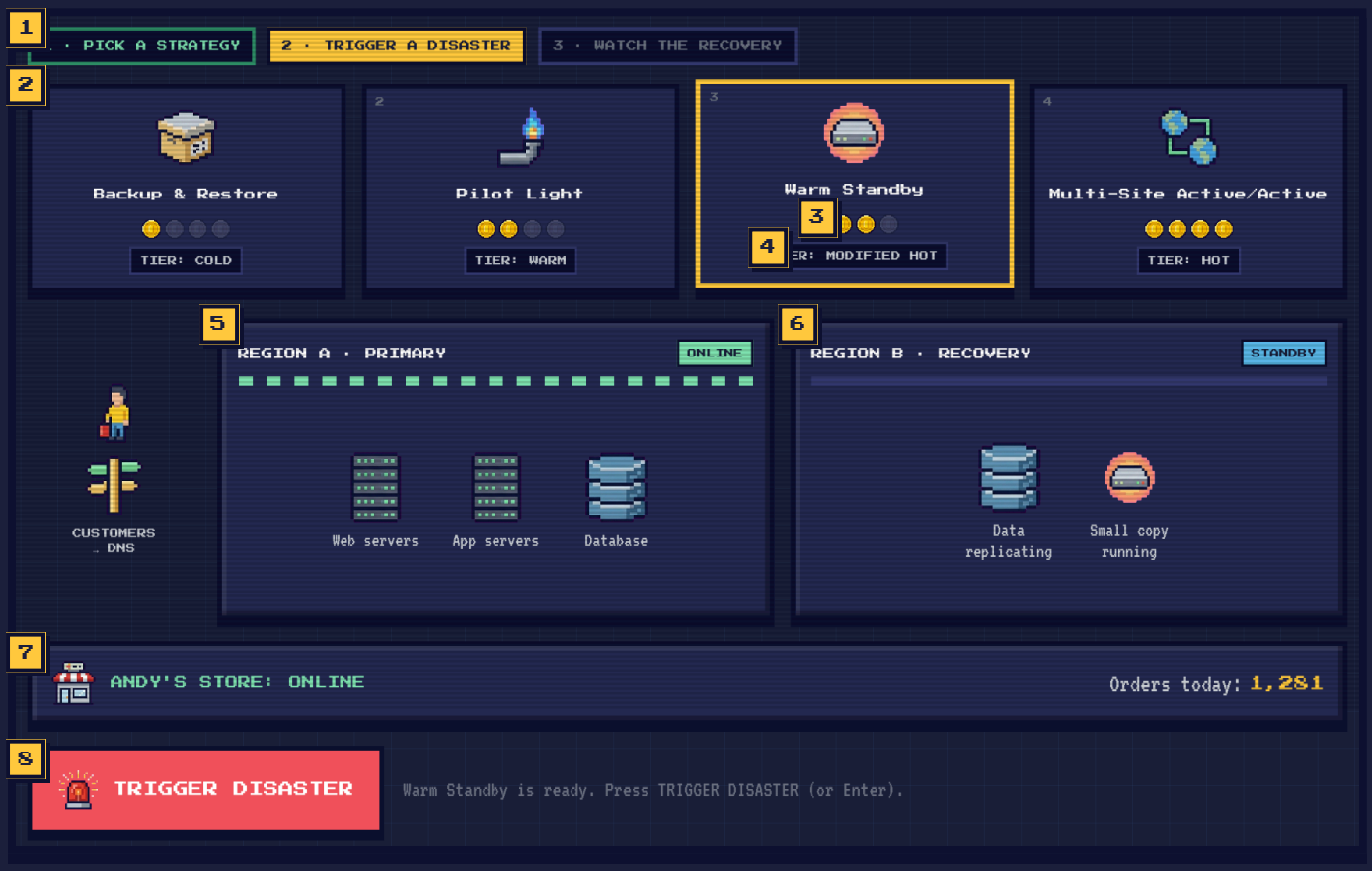
RTO · RECOVERY TIME OBJECTIVE

How long the store is down: the time from the disaster to full service again. Smaller is better.

RPO · RECOVERY POINT OBJECTIVE

How much data is lost, measured in time: anything saved after the last copy reached the backup site is gone. Smaller is better.

Screen tour



1. **Progress strip:** 1 pick a strategy, 2 trigger a disaster, 3 watch the recovery. The current step is yellow.
2. **Strategy cards** (keys 1–4). A green check appears on cards you've already tested.
3. **Monthly cost** in coins, from 1 (cheapest) to 4 (most expensive).
4. **Tier:** the closest backup tier from the report's Implementation Tiers table (Cold, Warm, Modified Hot, Hot).
5. **Region A:** Andy's main site, with web servers, app servers and the database.

6 Region B: the recovery site. What runs here depends on the strategy you pick.

7 Store status and the orders today counter, which goes up while the store is online.

8 TRIGGER DISASTER (ENTER): unlocks after you pick a strategy.

The four strategies

STRATEGY	WHAT'S IN REGION B BEFOREHAND	RTO	RPO	COST
Backup & Restore	Only backups. Servers have to be built from scratch.	8 h	4 h	1/4
Pilot Light	Data replicating, app servers switched off.	40 min	10 min	2/4
Warm Standby	Data replicating plus a small working copy of the store.	10 min	1 min	3/4
Multi-Site Active/Active	A full second site already serving customers.	30 s	about 3 s	4/4

Illustrative numbers. The order (slowest and cheapest to fastest and most expensive) follows the AWS Disaster Recovery whitepaper (2021).

During the disaster



1 Region A goes DOWN. The disaster changes every time: a typhoon, cut power lines or a security breach.

2 Recovery steps: the first step ("Disaster detected") shows red. Each later step lights up yellow while it runs and turns into a green check when done. Strategies that keep more running in Region B have fewer steps.

3 RTO clock: counts the downtime in game time (sped up). It turns green when the store is back.

4 RPO bar: how much recent data was lost. Long red bar = lots of lost orders.

The result



1 **Result:** downtime (RTO), data lost (RPO), orders affected and monthly cost. "Orders lost + missed" assumes 30 orders per hour during the holiday sale: orders lost from the RPO window plus orders missed while the store was down.

2 **Andy's reaction** to how it went.

3 **TRY ANOTHER STRATEGY** picks the next untested one. **BACK TO ZONES** returns to the hub.

4 **Compare your runs:** appears after 2 tested strategies. It clears when the page is reloaded.

5 After all 4, Andy's **guide question** appears. This is the class discussion question.

Walkthrough

RUN 1 The cheapest plan

1. Press **1** for **Backup & Restore**. Region B shows only backups.
2. Press **ENTER** (or click **TRIGGER DISASTER**).
3. Watch all five recovery steps. The clock runs to 8 hours and the RPO bar fills up: 4 hours of orders are gone.

RUN 2 The most expensive plan

1. Press `4` (or click the **Multi-Site Active/Active** card). Region B is already online.
2. Trigger the disaster. The store never goes dark and there are only three quick steps.
3. Look at the cost: 4 coins.

RUN 3 Answer Andy's question

1. Test **Pilot Light** (`2`) and **Warm Standby** (`3`) too.
2. Use the comparison table and Andy's situation (mid-sized, budget-conscious, small IT team, seasonal spikes) to decide which strategy you would recommend.
3. There is no single "correct" answer. Be ready to explain the trade-off you chose.

7 GLOSSARY

Availability
The share of time (or of requests) a service works. 99.999% is called "five nines" and allows less than 5.25 minutes of downtime a year (Endo et al., 2016).

Availability zone
A separate data center (with its own power, cooling and network) inside one cloud region.

Region
A geographic area with one or more data centers. Disaster recovery usually uses a second region.

Load balancer
The traffic director that spreads requests across servers and avoids unhealthy ones.

Health check
A regular "are you alive?" ping from the load balancer to each server.

Failover
Switching work from a failed component to a backup.

Redundancy
Keeping extra components so one failure doesn't stop the service.

2N / N+M / N-way active
Redundancy models: one full standby per active unit / M spares shared by N active units / every unit active and sharing the load.

Standby / warm pool
A backup server that is ready but not serving traffic yet. A warm pool is a group of pre-started spares.

Fault → error → failure
A fault (an underlying problem) causes an error (a wrong internal state), which can become a failure (something the user notices).

Proactive FT
Fault tolerance that predicts and prevents failures before they happen.

Reactive FT
Fault tolerance that responds after a failure has happened.

Checkpoint
A saved snapshot of a job's progress, so it can resume from there.

Replication
Running copies of a job or data in more than one place.

Disaster recovery
Preparing for and recovering from an event that stops a workload at its main location.

RTO

Recovery Time Objective: the maximum acceptable time between the interruption and the restoration of service.

RPO

Recovery Point Objective: the maximum acceptable data loss, measured in time.

DNS

The internet's address book. Switching DNS points customers to the recovery region.

8 TROUBLESHOOTING AND FAQ

? I can't hear anything.

Sound is off by default. Click **BGM** (music) and/or **SFX** (effects) in the top bar so they read ON. Also check your device volume and that the browser tab isn't muted.

? The music didn't start when I opened the page.

That's expected: sound is always off when the app opens. Click **BGM** once and the music starts right away.

? Something looks stuck.

HA zone: press **REPAIR ALL** or **RESET**. FT zone: press **RESTART**. DR zone: pick a strategy card again. You can always press `ESC` to go back to the zone screen and come back in.

? A button doesn't respond.

Dimmed buttons are locked on purpose. In the FT zone, proactive techniques unlock at 30% temperature and reactive ones after the crash. In the DR zone, TRIGGER DISASTER unlocks after you pick a strategy.

? My comparison tables disappeared.

Results are kept only while the page stays open. Reloading the page clears them.

? Why do the numbers drop so fast?

Everything is sped up so an outage fits into a few seconds of play. A real 99.9% service is down for minutes a month, not seconds.

? Are these real AWS or Netflix numbers?

No. The models, techniques and strategies are real and come from the sources, but every number in the game is illustrative.

? Does it work without internet?

Yes. Once the page has loaded, or if you open the project folder's `index.html`, everything (icons, fonts, music) is stored locally.

9 SOURCES AND CREDITS

Sources behind the concepts

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Credits

- Music: "Link to the Castle" by TAD ON (royalty free, YouTube).
- Fonts: Press Start 2P and VT323 (SIL Open Font License).
- Icons: pixel art generated for this project.



THANKS FOR PLAYING!

Uptime Quest and this manual were made by **Claude L. Calimlim**.